Formal Derivation Structure of the Scalar–Angular–Torsion Framework

Foundational Structure

A formalization of the Scalar–Angular–Torsion (SAT) framework, grounded in the integrated four-dimensional Hyperhelical (4DHH) and Hypersphere Unit Cell (UC) action, yields a hierarchical derivation structure. The construction proceeds from fundamental four-dimensional geometric primitives to emergent macroscopic physics.

Root: Unified Blockwave Action

The global action integrates discrete 24-cell hyperspherical lattice dynamics with continuous four-dimensional hyperhelical kinematics:

$$S_{\text{Total}} = \int d^4x \, \sqrt{-g} \, (\mathcal{L}_{\text{UC}} + \mathcal{L}_{\text{4DHH}}) \, . \tag{1}$$

I. Gravitational Sector

Metric Emergence

The spacetime metric is defined as the inverse of the ensemble average of filament tangent vectors:

$$g_{\mu\nu} \equiv (\langle v_\mu v_\nu \rangle)^{-1} \, . \tag{2}$$

Einstein–Hilbert Block

General Relativity is recovered through a macroscopic curvature term generated by filament ensemble distortion:

$$S_{\text{GR}} = \frac{1}{16\pi G} \int d^4x \, \sqrt{-g} \, R. \tag{3}$$

The gravitational coupling satisfies the retrodicted relation

$$\frac{G}{c^4} \to 8\pi \ell_f^2. \tag{4}$$

A higher-curvature correction appears with fixed coefficient

$$\beta = \frac{2}{6\pi} \approx 0.106, \tag{5}$$

yielding an additional contribution

$$S_{R^2} = \beta \int d^4x \sqrt{-g}\, R^2. \tag{6}$$

Newtonian Limit

In the weak-field regime, the Poisson equation is recovered:

$$\nabla^2 \Phi = 4\pi G \rho. \tag{7}$$

Test particle motion follows from the geodesic limit of the weak-field metric derived from S_{Total}.

II. Interaction and Gauge Sector

Topological Mode Densities

Interaction strengths g_G arise from statistical frequencies of lattice linking configurations.

The fine-structure constant is expressed through $U(1)$ winding mode densities and a projection constant

$$B = \frac{3}{4\pi}. \tag{8}$$

Gauge Group Emergence

The following gauge structures emerge:

$$U(1) : \text{single-filament phase twists,} \tag{9}$$
$$SU(2) : \text{two-filament Hopf exchange redundancies,} \tag{10}$$
$$SU(3) : \text{Borromean triplet phase permutations.} \tag{11}$$

Renormalization Structure

Ultraviolet completion is ensured by a constraint

$$Q \leq 3, \tag{12}$$

with divergences absorbed into two renormalization factors,

$$Z_F, \qquad Z_\Sigma, \tag{13}$$

at all loop orders.

III. Matter Sector

Integrated Geometric Invariants

Particle properties arise as integrated invariants of four-dimensional worldline coiling histories.

Proportional Mass Law

Rest mass is proportional to topological charge:

$$m \propto Q. \tag{14}$$

Fermion Emergence

The Dirac equation emerges from normalized tangent vectors v^μ of linked filament bundles:

$$(i\gamma^\mu \nabla_\mu - m)\psi = 0. \tag{15}$$

Flavor structure is associated with discrete phase factors of the alternating group A_4.

Charge Sectors

Leptonic states correspond to $Q = 1$ configurations. Baryonic states correspond to $Q = 3$ Borromean triplets.

IV. Thermodynamic and Singular Sector

Thermodynamic Mapping

Four-dimensional hyperhelical dynamics map onto the Gibbs free energy identity:

$$G = U + pV - TS. \tag{16}$$

The vibrational spectrum is characterized by a Debye-type dispersion with sound speed v_s.

Black Hole Quantization

Area quantization follows from integer flux of a three-form current J:

$$S = n. \tag{17}$$

Discrete evaporation proceeds via transitions

$$n \to n - 1. \tag{18}$$

V. Cosmological Sector

Expansion Dynamics

Cosmic expansion is attributed to structural resolution within the filament manifold.

Inflation corresponds to a transient enhancement phase. Large-scale structure observables such as H_0 and S_8 arise from geometric saturation of the expanding time surface.

CMB Birefringence

Polarization rotation is predicted as

$$\Delta\theta = 0.30^\circ, \tag{19}$$

emerging from projection geometry with projection constant $B = 3/(4\pi)$.